

Summary Statistics

For dataset A vs C

"filtered genes: 5975"

"significant genes: 277"

"upregulated genes: 144"

"downregulated genes: 133"

"summary of pvalues:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03189	0.21811	0.32323	0.57588	1.00000

"summary of log2FoldChange:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-2.363e+01	-2.233e-01	-8.733e-04	-2.374e-02	2.219e-01	2.028e+01

For Dataset A vs D

"filtered genes: 6094"

"significant genes: 1016"

"upregulated genes: 365"

"downregulated genes: 651"

"summary of pvalues:"

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001665	0.074704	0.234986	0.415297	1.000000

"summary of log2Foldchange:"

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-25.328550	-0.503281	-0.003507	0.105492	0.440455	22.002860

Volcano Plots

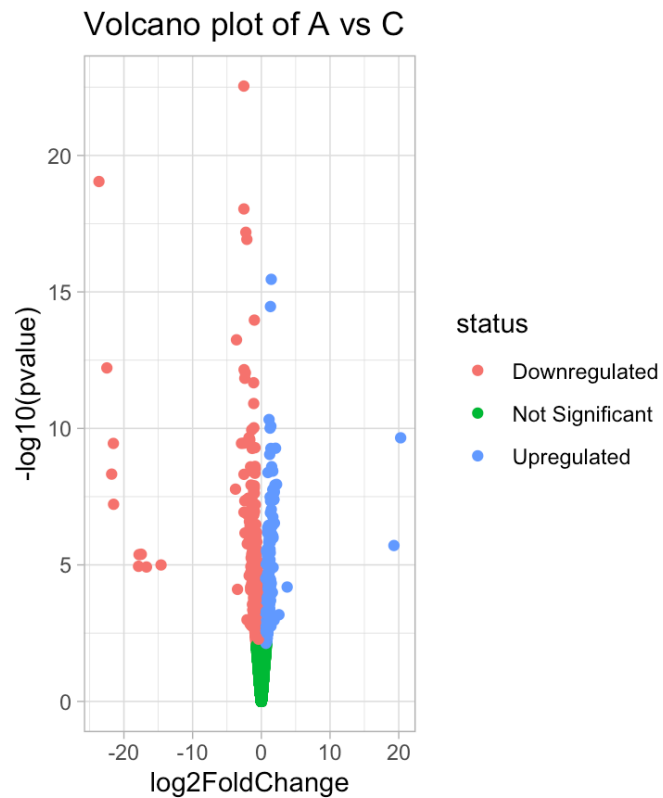


Fig 1.The above volcano plot compares log2FoldChange with -log10(pvalue) for dataset A vs C. The genes are higher as compared to the other dataset which shows greater significance.

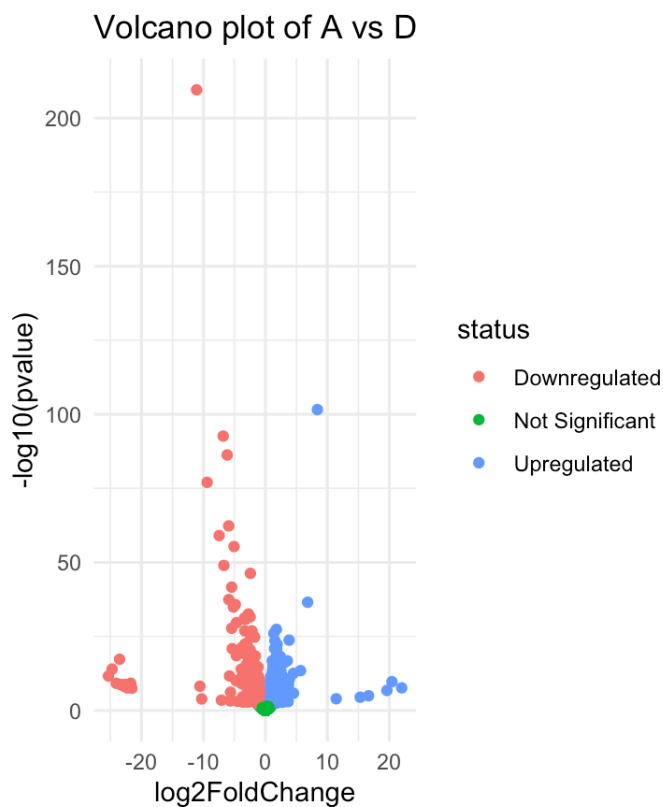


Fig 2.The above volcano plot compares log2FoldChange with -log10(pvalue) for dataset A vs D.

MA Plots

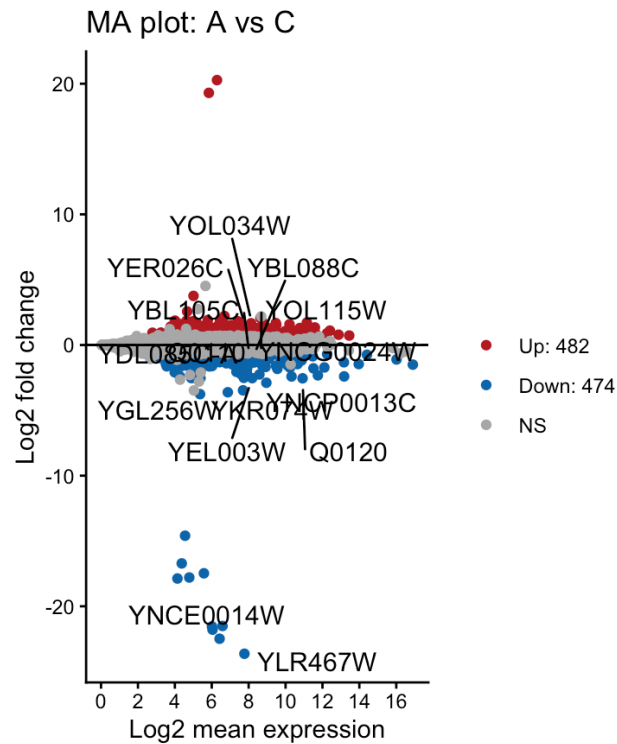


Fig 3.The above MA plot compares log2 mean expression with log2FoldChange for dataset A vs C.

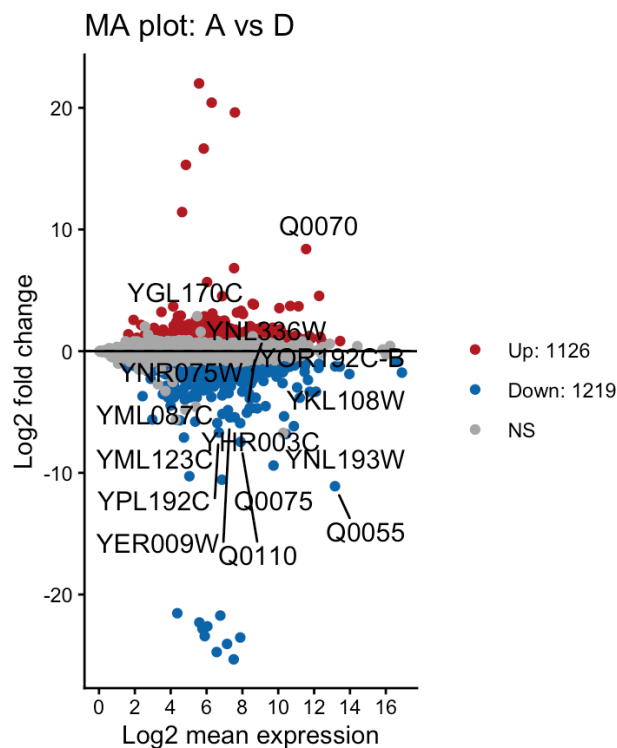


Fig 4.The above MA plot compares log2 mean expression with log2FoldChange for dataset A vs D.

Histograms

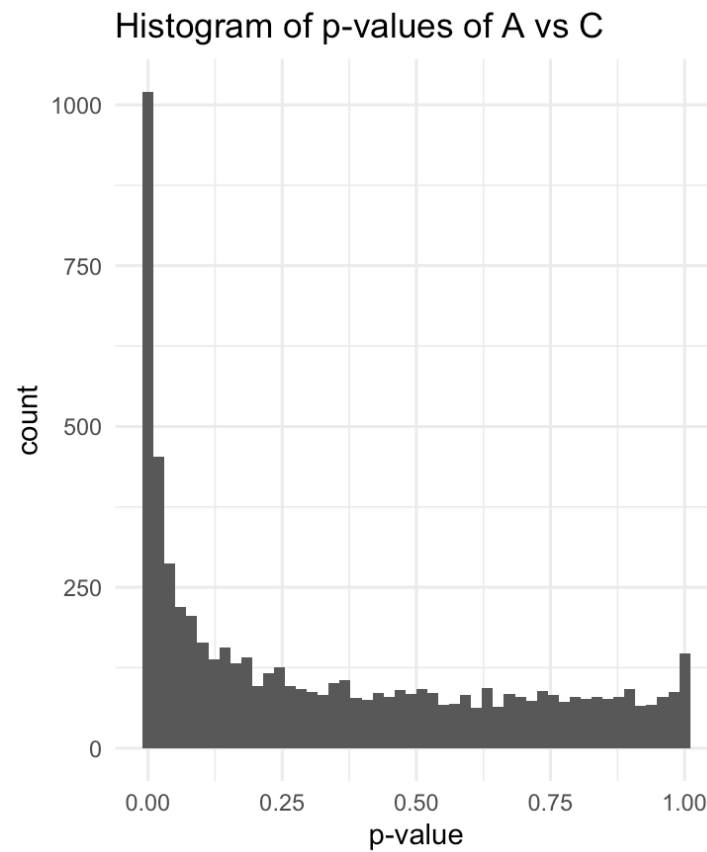


Fig 5.The above is a histogram of p-values of dataset A vs C. The peak near 0.00 shows that many genes are differentially expressed.

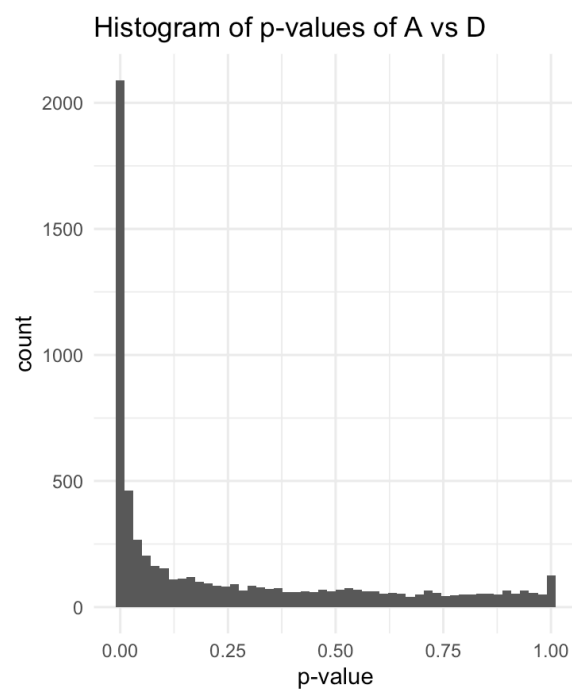


Fig 6.The above is a histogram of p-values of dataset A vs D. Spike near zero suggests many significant genes.

Heatmaps

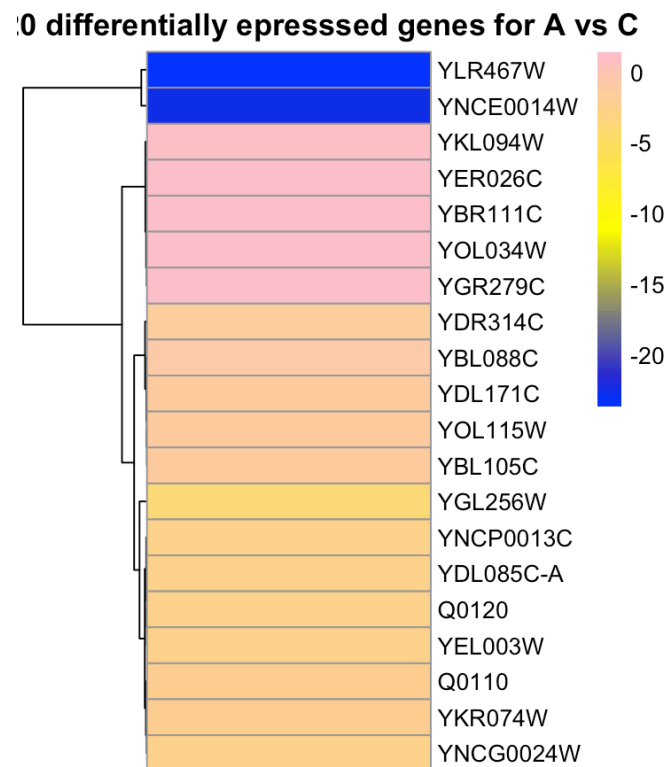


Fig 7. The above heatmap visualizes log2FoldChange of top 20 differentially expressed genes in dataset A vs C.

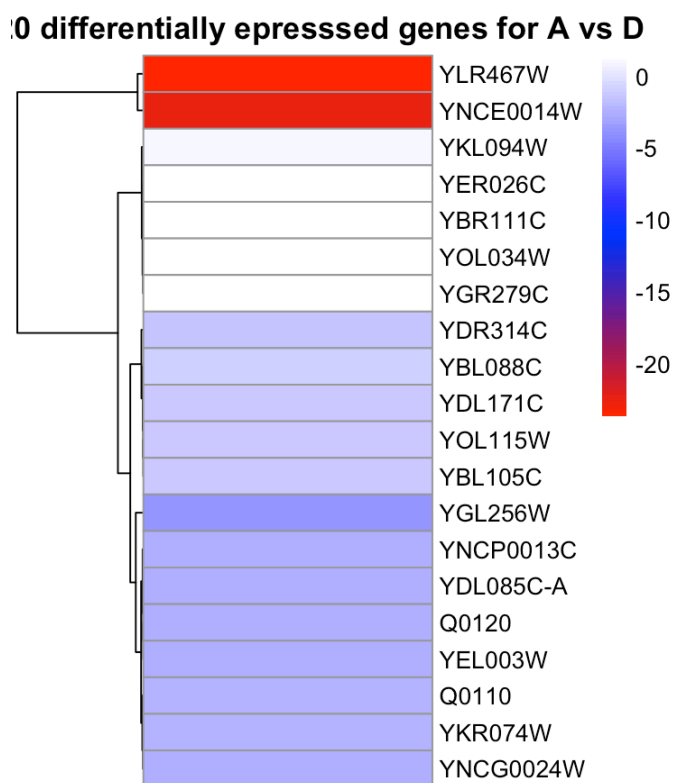


Fig 8. The above heatmap visualizes log2FoldChange of top 20 differentially expressed genes in dataset A vs D.

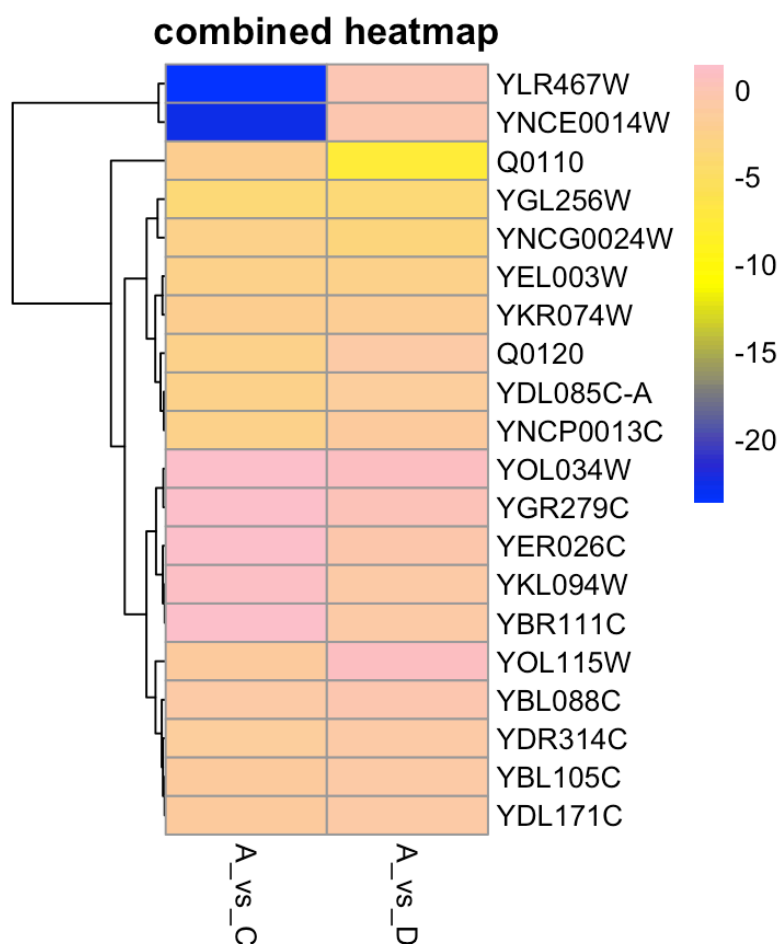


Fig 9.The above heatmap compares log2FoldChange of top 20 differentially expressed genes in dataset A vs C and dataset A vs D.

Top few lines significantly upregulated genes of dataset A vs C

gene_id	log2FoldChange	pvalue	padj
YOL034W	1.442559	3.457892e-16	3.443484e-13
YER026C	1.327270	3.433684e-15	2.930895e-12
YKL094W	1.123977	4.727288e-11	1.765346e-08
YGR279C	1.385148	8.273279e-11	2.907814e-08
YBR111C	1.272053	9.764574e-11	3.070702e-08
YNCJ0019C	20.276410	2.208957e-10	5.999325e-08

Top few lines significantly downregulated genes of dataset A vs C

gene_id	log2FoldChange	pvalue	padj
Q0120	-2.550839	2.879866e-23	1.720720e-19
YLR467W	-23.634090	9.026527e-20	2.696675e-16
YEL003W	-2.536806	9.120245e-19	1.816449e-15
YKR074W	-2.254309	6.535487e-18	9.762383e-15
Q0110	-2.106980	1.187316e-17	1.418842e-14
YBL088C	-1.014724	1.076490e-14	8.040034e-12

Top few lines significantly upregulated genes of dataset A vs D

gene_id	log2FoldChange	pvalue	padj
Q0070	8.39	2.66e-102	8.10e-99
YGL170C	6.82	2.94e- 37	1.38e-34
YNR022C	1.78	4.06e- 28	1.12e-25
YGR184C	1.37	9.31e- 27	2.18e-24
YIR019C	3.83	1.80e- 24	3.78e-22
YOR017W	1.53	2.89e- 24	5.88e-22

Top few lines significantly downregulated genes of dataset A vs D

gene_id	log2FoldChange	pvalue	padj
Q0055	-11.1	2.89e-210	1.76e-206
YNL193W	-6.81	2.33e- 93	4.74e- 90
YKL108W	-6.16	5.42e- 87	8.26e- 84
Q0075	-9.40	9.46e- 78	1.15e- 74
YHR003C	-5.92	4.80e- 63	4.87e- 60
Q0110	-7.46	8.92e- 60	7.76e- 57

Additional Analysis

Clustering of gene expression patterns

a hierarchal clustering heatmap was already generated in the heatmap section that had the top 20 differentially expressed genes common between both datasets

